DFIR Evidence Register & Timeline

Use this workbook to learn a repeatable method. Write what you observed separately from what you infer; never present fictional sample data as a real incident or production assessment.

1. Establish scope first

SAFETY BOUNDARY This lab uses fictional records and a hypothetical case. Do not acquire, inspect, alter, or upload someone else's files or devices. Work only with supplied evidence or material you are explicitly authorised to handle.

Case field	Learner entry
Case ID / purpose	
Authorised scope and exclusions	
Time window and time zone	
Evidence source / supplied filename	
Analyst and review date	
Known constraints or missing data	

2. Evidence register

Record provenance before interpretation. Use one row per evidence item; preserve supplied source files and describe any working copy or transformation.

Evidence ID	Source / reference	Collected / received (UTC)	Integrity / handling note
EV-001			
EV-002			
EV-003			
EV-004			

Handling checklist

- ☐ Scope and authority are recorded before analysis.
- ☐ Original supplied files are preserved; analysis notes identify the copy used.
- ☐ Sensitive identifiers are redacted before sharing.
- ☐ Time zone and any clock uncertainty are recorded.

3. Build a defensible timeline

Use UTC when possible. Keep the original timestamp and record any normalisation. A timestamp match alone does not prove two events share a cause.

Time (UTC)	Evidence ID / source	Observed fact	Possible interpretation	Unknown / next question

4. Separate facts from hypotheses

Observed fact (with source)	Hypothesis / explanation	Confidence and reason	Evidence needed to verify

5. Close with an honest summary

What is supported by the supplied evidence?

What remains unknown or ambiguous?

What safe next check would reduce uncertainty?

QUALITY CHECK Can another learner trace every conclusion back to a source row? If not, add the missing reference or weaken the claim.

A real lab for these notes

Topic: Evidence preservation, file integrity, timeline reconstruction, and careful reporting. These pages connect the existing OpenCyberPath notes to specific provider-hosted exercises. The card label is intentionally generic; the exact task name and direct URL are listed here so you can open the right exercise instead of a whole catalogue.

Concept checkpoints

- Preserve original evidence; analyze a verified copy.
- Record source, collector, UTC time, tool/version, SHA-256, storage path, and every transfer.
- A timeline event is not automatically a malicious action; corroborate it with other sources.
- Fictional CSV fields are training records, not a chain of custody for a real case.

Autopsy *[TryHackMe]*
https://tryhackme.com/room/btautopsye0
Access note: Account required; the room describes a deployable lab machine and a disk-image investigation. Some tasks may be limited by the current plan.

Linux Logging for SOC *[TryHackMe]*
https://tryhackme.com/room/linuxloggingforsoc
Access note: Account required; the room provides a lab VM for examining Linux system and authentication logs.

ResourcePacks *[CyberDefenders]*
https://cyberdefenders.org/blueteam-ctf-challenges/resourcepacks/
Access note: Direct specific challenge page for timeline reconstruction. This challenge is marked retired and requires sign-in to download evidence; use the interactive Autopsy room above if unavailable.

ACCESS CHECK Open the exact task page and verify the current login, account tier, VM launch button, free/paid terms, and any cost notes before starting. A provider may change availability or retire a challenge.

Before you click Start

- Write down the provider's target/scope and use only the instance it creates for you.
- Keep this notebook beside the exercise. Use its original course data before trying commands on a local machine.
- Capture timestamps, filenames, row/packet IDs, and a short explanation; redact credentials, tokens, and personal data.
- If a page says retired, premium, or downloadable-only, use the alternative listed in the access note or return to the matching interactive lab above.

Commands you can explain

Only analyze the sample files supplied by the course or evidence you are formally authorised to handle. Do not inspect someone else's device or files.

First download the matching sample file(s) from the existing card when a command names one. Keep the data in a practice folder and never execute strings stored in evidence.

WINDOWS / HASH AN AUTHORISED SAMPLE

```
Get-FileHash .\evidence-sample.bin -Algorithm SHA256
```

Replace the file name with an assigned evidence copy; record the hash before analysis.

LINUX / HASH AN AUTHORISED SAMPLE

```
sha256sum ./evidence-sample.bin
```

Creates a checksum string without changing the evidence file.

POWERSHELL / LIST METADATA

```
Get-ChildItem .\evidence -File | Select-Object Name,Length,LastWriteTimeUtc
```

Inventory files in the assigned evidence-copy directory. Timestamps may be modified by copying and are not conclusive alone.

LINUX / SEARCH SUPPLIED LOG COPY

```
grep -niE "failed|accepted|sudo|session" ./case/auth.log | head -n 50
```

Search a copy of the training log. Keep the source file untouched and note the matching line numbers.

LINUX / FILE-TYPE CHECK

```
file ./evidence-sample.bin
```

Identify a file type from its contents; this is not proof the file is safe to execute.

COMMAND HABIT Check the current working directory, shell, target, and expected effect before pressing Enter. Save the exact command and the output that mattered.

Guided lab run

- 1. Open the exact Autopsy lab and start the attached VM; alternatively use the assigned forensic case files.
- 2. Record case ID, authority/scope, evidence source, hash, system timezone, and working-copy path.
- 3. Build a timeline in UTC and keep source record IDs alongside parsed timestamps.
- 4. Separate facts from hypotheses and mark missing data or timestamp uncertainty.
- 5. Export a timeline and concise summary; never upload actual third-party evidence to a public service.

Evidence checklist

- [ ] What exact lab instance or sample file was used?
- [ ] Which command/query and time range produced the observation?
- [ ] What is the directly observed fact, separate from your inference?
- [ ] What benign/alternative explanation could fit?
- [ ] What extra record or control would help confirm the conclusion?

Check your understanding

Q What does a matching SHA-256 hash show?

Think it through: It supports that two byte sequences match; it does not prove the source is authentic or the file is harmless.

Q Why normalise timestamps to UTC?

Think it through: It permits comparison across log sources and time zones while retaining the original timestamp and offset.

Q What is chain of custody?

Think it through: A documented record of who handled evidence, when, why, and how integrity was preserved.

NOTEBOOK CLOSE-OUT Finish with: Observation / Evidence source / Why it matters / Alternative explanation / Next check / Limitation. A careful "not enough evidence yet" is a valid professional result.